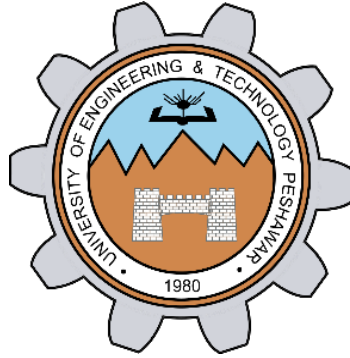




**UNIVERSITY OF ENGINEERING AND TECHNOLOGY PESHAWAR
(NOWSHERA CAMPUS)**

**COMPUTER
PROGRAMMING**

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Lab Report

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Program No: 1

```
#include <iostream>
using namespace std;
```

```

int main() {
int arr[10], max, min, sum = 0;

// Taking input from user
cout << "Enter 10 integers:\n";
for(int i = 0; i < 10; i++) {
cin >> arr[i];
}

// Initialize max and min with first element
max = min = arr[0];

// Loop to find max, min and sum
for(int i = 0; i < 10; i++) {
if(arr[i] > max) max = arr[i]; // Update max
if(arr[i] < min) min = arr[i]; // Update min
sum += arr[i]; // Add to sum
}

// Display results
cout << "Maximum: " << max << endl;
cout << "Minimum: " << min << endl;
cout << "Average: " << sum / 10.0 << endl;

return 0;
}

```

Program No: 2

```

#include <iostream>
using namespace std;

// Function to reverse array
void reverseArray(int arr[], int size) {
for(int i = 0; i < size/2; i++) {
swap(arr[i], arr[size - i - 1]); // Swap elements
}
}

int main() {
int arr[5] = {1,2,3,4,5};

```

// Display original array

```
cout << "Before Reversal: ";  
for(int i=0;i<5;i++) cout << arr[i] << " ";
```

reverseArray(arr, 5); **// Function call**

// Display reversed array

```
cout << "\nAfter Reversal: ";  
for(int i=0;i<5;i++) cout << arr[i] << " ";
```

```
return 0;  
}
```

Program No: 3

```
#include <iostream>  
using namespace std;
```

```
int main() {  
int A[3][3], B[3][3], C[3][3];
```

// Input Matrix A

```
cout << "Enter Matrix A:\n";  
for(int i=0;i<3;i++)  
for(int j=0;j<3;j++)  
cin >> A[i][j];
```

// Input Matrix B

```
cout << "Enter Matrix B:\n";  
for(int i=0;i<3;i++)  
for(int j=0;j<3;j++)  
cin >> B[i][j];
```

// Add matrices

```
for(int i=0;i<3;i++)  
for(int j=0;j<3;j++)  
C[i][j] = A[i][j] + B[i][j];
```

// Display result

```
cout << "Result Matrix:\n";  
for(int i=0;i<3;i++) {
```

```
for(int j=0;j<3;j++)  
cout << C[i][j] << " ";  
cout << endl;  
}
```

```
return 0;  
}
```

Program No: 4

```
#include <iostream>  
using namespace std;
```

```
int main() {  
string names[5];
```

// Input names

```
cout << "Enter 5 names:\n";  
for(int i=0;i<5;i++)  
cin >> names[i];
```

// Sorting using simple comparison

```
for(int i=0;i<5;i++) {  
for(int j=i+1;j<5;j++) {  
if(names[i] > names[j]) {  
swap(names[i], names[j]); // Swap names  
}  
}  
}
```

// Display sorted names

```
cout << "Sorted Names:\n";  
for(int i=0;i<5;i++)  
cout << names[i] << endl;
```

```
return 0;  
}
```

Program No: 5

```
#include <iostream>  
using namespace std;
```

// Function to calculate total load

```
double totalLoad(double arr[], int size) {  
    double sum = 0;  
    for(int i=0;i<size;i++) sum += arr[i];  
    return sum;  
}
```

// Function to find peak load

```
double peakLoad(double arr[], int size) {  
    double max = arr[0];  
    for(int i=1;i<size;i++)  
        if(arr[i] > max) max = arr[i];  
    return max;  
}
```

```
int main() {  
    double load[24];
```

// Input load data

```
    cout << "Enter 24-hour load:\n";  
    for(int i=0;i<24;i++)  
        cin >> load[i];
```

// Display results

```
    cout << "Total Load: " << totalLoad(load,24) << endl;  
    cout << "Peak Load: " << peakLoad(load,24) << endl;
```

```
    return 0;  
}
```

Program No: 6

```
#include <iostream>  
using namespace std;
```

// Function to find topper

```
void topper(string names[], int marks[], int size) {  
    int max = marks[0], index = 0;  
    for(int i=1;i<size;i++) {  
        if(marks[i] > max) {  
            max = marks[i];  
            index = i;
```

```

    }
}
cout << "Topper: " << names[index] << endl;
}

// Function to display pass/fail
void passFail(string names[], int marks[], int size) {
for(int i=0;i<size;i++) {
cout << names[i] << " - ";
if(marks[i] >= 50) cout << "Pass\n";
else cout << "Fail\n";
}
}

```

```

// Function to search student
void search(string names[], int size, string key) {
for(int i=0;i<size;i++) {
if(names[i] == key) {
cout << "Found: " << names[i] << endl;
return;
}
}
cout << "Not Found\n";
}

```

```

int main() {
string names[3] = {"Ali", "Ahmed", "Sara"};
int marks[3] = {80, 45, 90};

topper(names, marks, 3);
passFail(names, marks, 3);
search(names, 3, "Ahmed");

return 0;
}

```

Program No: 7

```

#include <iostream>
using namespace std;

```

// Calculate total energy

```
double totalEnergy(double arr[]) {  
    double sum=0;  
    for(int i=0;i<24;i++) sum+=arr[i];  
    return sum;  
}
```

// Find hour of maximum generation

```
int maxHour(double arr[]) {  
    int index=0;  
    for(int i=1;i<24;i++)  
        if(arr[i] > arr[index]) index = i;  
    return index;  
}
```

```
int main() {  
    double energy[24];
```

// Input data

```
for(int i=0;i<24;i++)  
    cin >> energy[i];
```

```
cout << "Total Energy: " << totalEnergy(energy) << endl;  
cout << "Max Hour: " << maxHour(energy) << endl;  
cout << "Average: " << totalEnergy(energy)/24 << endl;
```

```
return 0;  
}
```

Program No: 8

```
#include <iostream>  
using namespace std;
```

```
int main() {  
    double v[20];  
    int low=0, high=0;
```

// Input voltage readings

```
for(int i=0;i<20;i++) cin >> v[i];
```

// Check safe limits

```
for(int i=0;i<20;i++) {  
    if(v[i] < 3.0) low++;  
    if(v[i] > 4.2) high++;  
}
```

// Find max and min voltage

```
double max=v[0], min=v[0];  
for(int i=1;i<20;i++) {  
    if(v[i] > max) max=v[i];  
    if(v[i] < min) min=v[i];  
}
```

```
cout << "Low: " << low << endl;  
cout << "High: " << high << endl;  
cout << "Max: " << max << endl;  
cout << "Min: " << min << endl;
```

// Warning if unsafe

```
if(low>0 || high>0)  
    cout << "Warning: Unsafe readings!\n";  
  
return 0;  
}
```

Program No: 9

```
#include <iostream>  
using namespace std;
```

```
int main() {  
    int arr[4][7];
```

// Input data

```
for(int i=0;i<4;i++)  
    for(int j=0;j<7;j++)  
        cin >> arr[i][j];
```

// Calculate total per region

```
for(int i=0;i<4;i++) {  
    int sum=0;  
    for(int j=0;j<7;j++)
```



```
sum+=arr[i][j];
cout << "Region " << i+1 << " Total: " << sum << endl;
}
return 0;
}
```

Program No: 10

```
#include <iostream>
using namespace std;
```

```
int main() {
int att[5][7];
```

// Input attendance

```
for(int i=0;i<5;i++)
for(int j=0;j<7;j++)
cin >> att[i][j];
```

```
int max=0, index=0;
```

// Calculate attendance

```
for(int i=0;i<5;i++) {
int sum=0;
for(int j=0;j<7;j++)
sum+=att[i][j];
```

```
cout << "Student " << i+1 << ": " << sum << endl;
```

```
if(sum > max) {
max = sum;
index = i;
}
}
```

```
cout << "Highest Attendance: Student " << index+1 << endl;
return 0;
}
```